PUBLIC-DATA PORTFOLIO CASE STUDY

Demand planning for urban mobility

Pricing, credit economics and a holiday-week plan



Specialist – GIP Demand Planning

NYC · Uber & Lyft · 2024–2025 research data

Prepared October 2026 · Illustrative cover image

Three decisions supported by the analysis

Turn observed signals into a plan, a commercial hypothesis and a test.

Planning question	Evidence from the project	Recommended action
Where is demand shifting?	−9.49% relative zone growth gap; operator responses differ.	Monitor operator and trip segments before a pricing response.
Should credit be expanded?	+1.61 pp share difference; $25.88 scenario cost per extra trip.	Test targeted credit and incremental contribution before scale.
What volume should be planned?	4.871M planned trips; 3.48% retrospective WAPE.	Use the two-week weekday mean, with a holiday scenario.

Decision stance: plan the week, investigate the shift, test the spend.

A reproducible view of served demand

59.41M

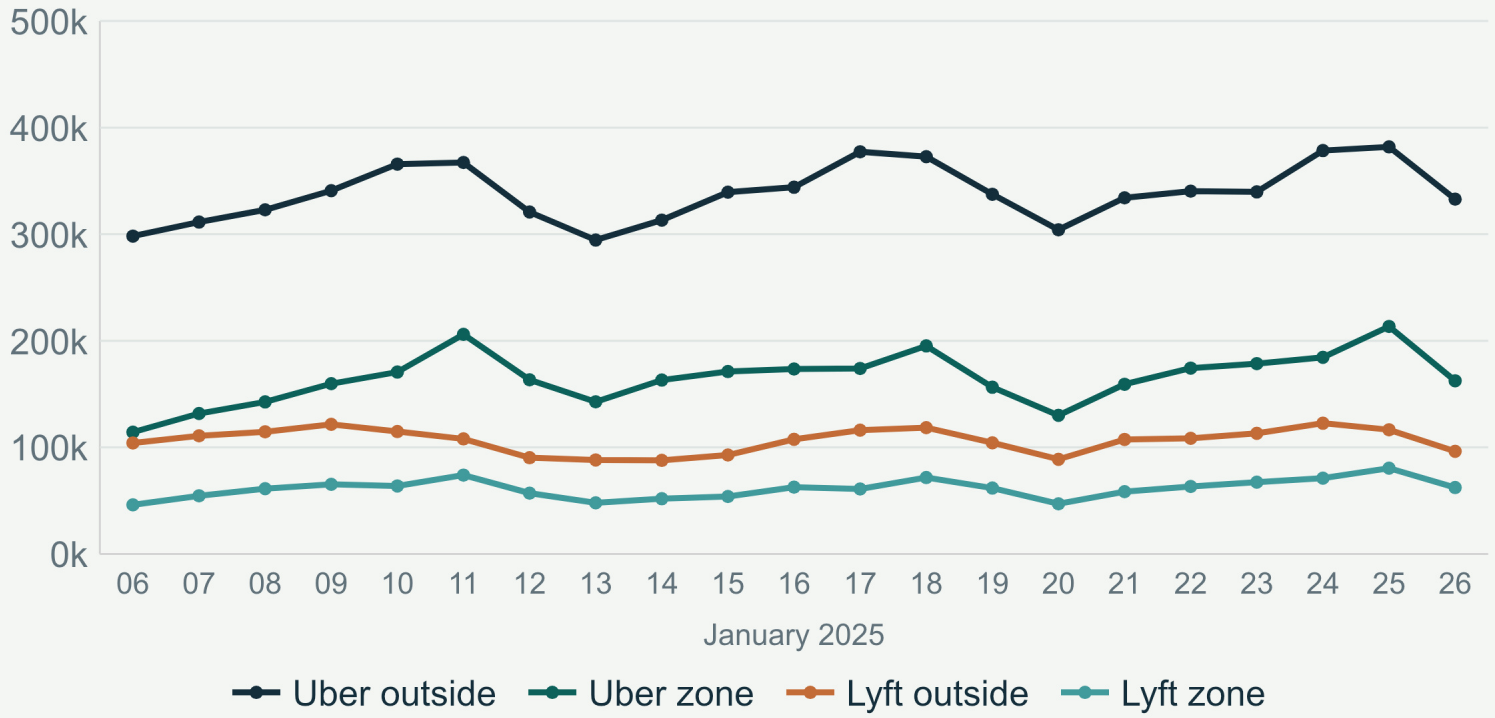
retained completed trips
Jan 2024 · Jan & Feb 2025

Work delivered	Definition and control
Source integrity	Immutable raw files, source URLs and SHA-256 checksums.
Segment logic	Uber / Lyft × zone / outside; 38 zones at the 95% threshold.
Metric quality	Keep trip volume; use separate valid samples for fare and wait.
Reproducibility	DuckDB; 360 daily rows; 2,160 hourly rows; independent notebook.

Separate rain study: 58.43M summer trips. Standalone Jupyter execution verified; actual Google Colab runtime remains untested.

Daily monitoring separates alerts from explanations

Daily completed trips by operator and endpoint-zone segment · 6–26 Jan 2025



118

alert rows across 58 days
in the full monitoring scope

Jan 21 · Lyft zone

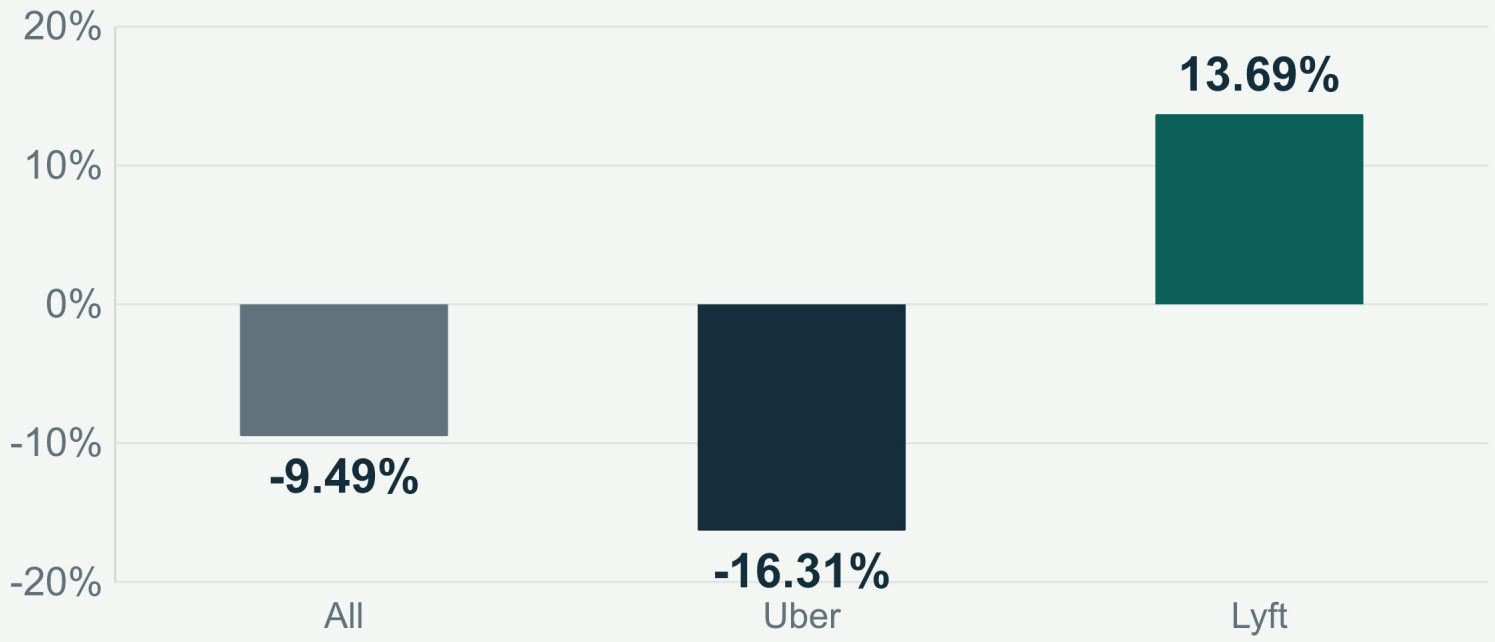
58,431 trips retained.
10.3% excluded from fare metrics.
Root cause remains unproven.

Watchlist covers the 90-day main scope; the chart is a 21-day excerpt. 56 rows lack a usable baseline; two cases were reviewed.

PRICING SIGNAL

Zone response differed sharply by operator

Relative zone growth gap, normalized to outside-zone growth · matched weekday windows



+1.97%

total market volume growth in the same comparison

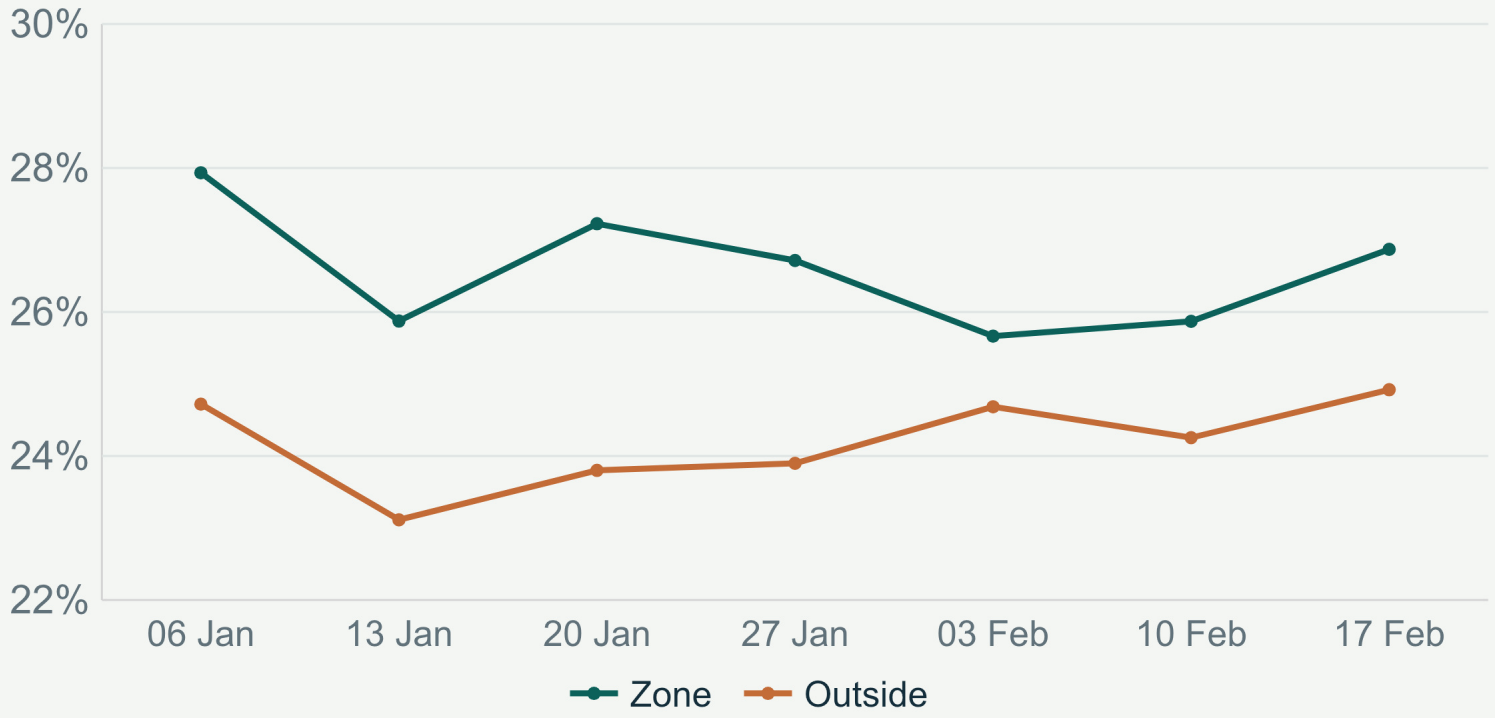
The segment mix matters.

Review short and low-fare trips; keep operator responses separate.

Jan 8–28, 2024 vs Jan 6–26, 2025. Direction survives zone-threshold and rain-window checks; causality is not identified.

Lyft share changed; incremental demand is uncertain

Weekly Lyft share of Uber + Lyft completed trips · complete observed weeks only



+1.61 pp

observed share difference between zone and outside

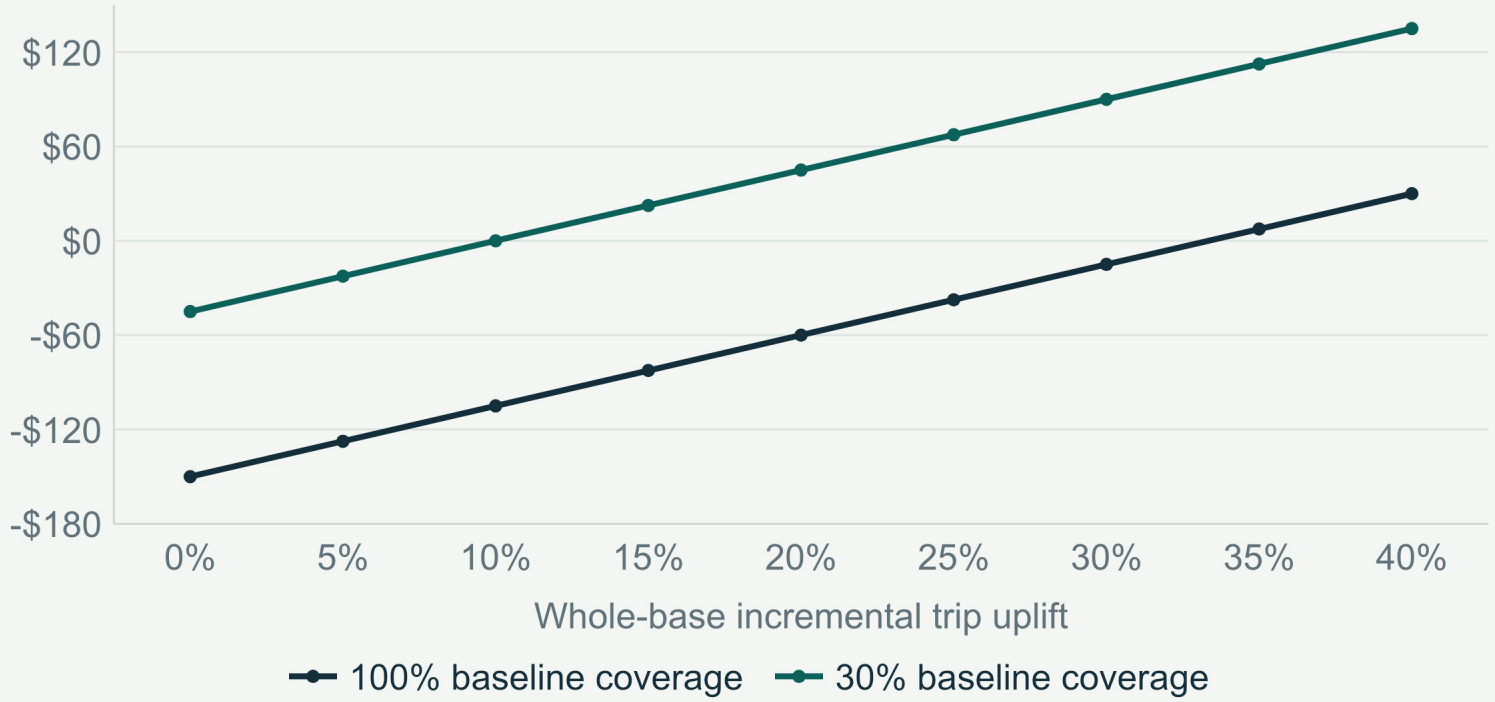
6.15% uplift proxy

Requires a fixed market and an unaffected comparison group.

Credit: $1.50 for future use, issued Jan 5–31 with a one-week validity. Comparison: Jan 6–26 vs Feb 3–23; carryover can affect controls.

Blanket credit is costly under the stated assumptions

Net incremental contribution per 100 baseline trips · scenario model, USD



$25.88

scenario cost per extra trip
at the 6.15% uplift proxy

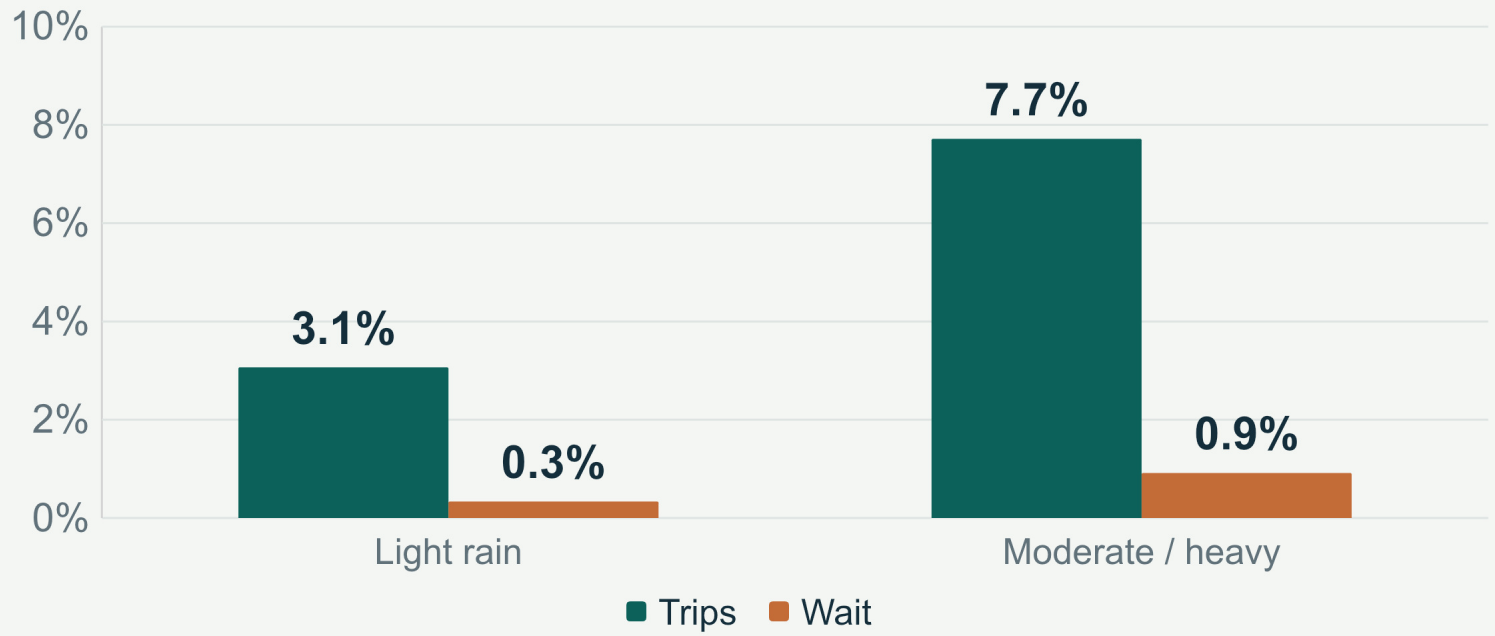
Break-even uplift

33.33% · blanket coverage
10.00% · 30% coverage

Assume $1.50 credit, $6 contribution, full redemption and credit on every extra trip. Targeted uplift must be re-estimated.

Use rain factors as monitoring triggers

Observed deviation from matched dry hours · June–August 2024



35 hours

usable moderate/heavy rain across 14 wet events

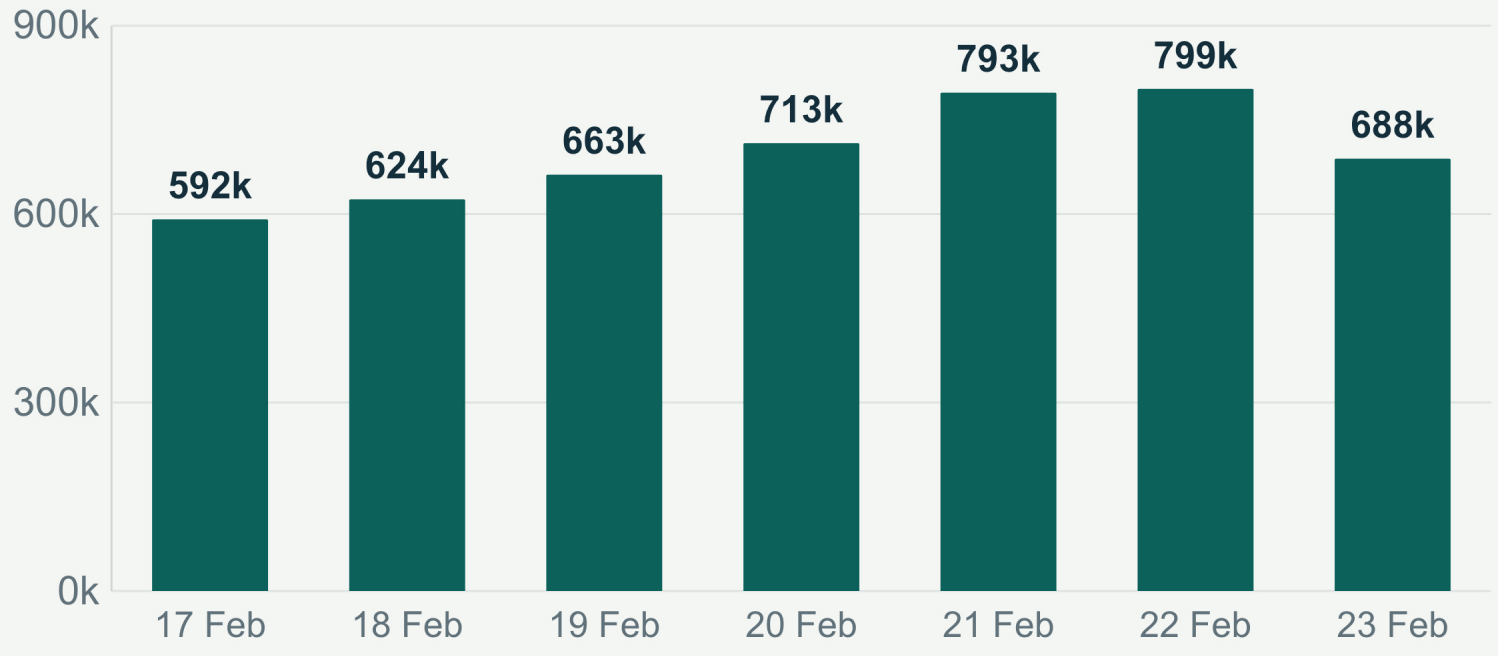
Use as a watch trigger.

Season, event clustering and city-wide weather limit transfer to winter.

110 usable light-rain hours. No weather adjustment was applied to the holiday-week forecast; completed trips do not measure supply.

A holiday-week plan using pre-cutoff data

Plan fixed at end of 16 Feb 2025, New York time · target 17–23 Feb



4.871M

planned completed trips across four daily segments

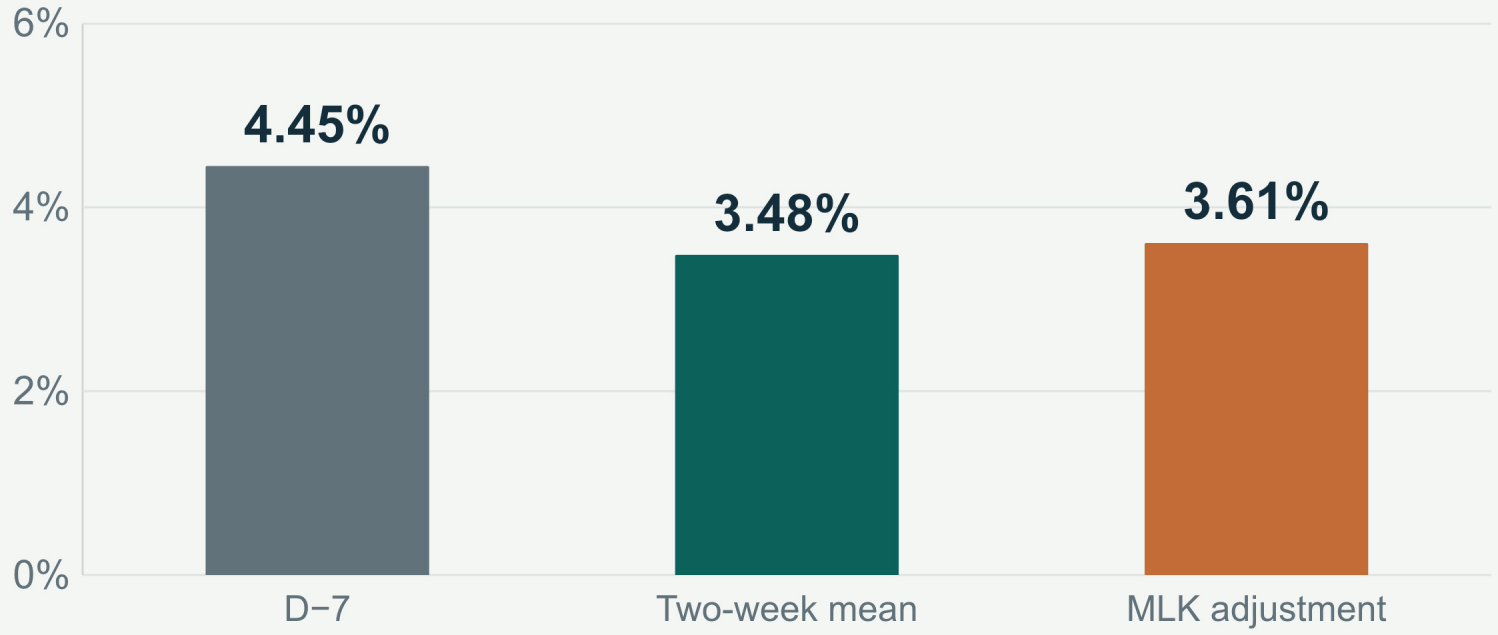
Baseline: mean of D−7 and D−14.

MLK transfer retained as a Feb 17 scenario.

9,961 post-cutoff completions excluded. Retrospective simulation assumes records were available; TLC monthly publication is delayed.

The two-week mean performed best in this test week

Weighted absolute percentage error · 28 daily segment forecasts



3.48%

WAPE for the two-week mean with +1.06% signed bias

A provisional baseline.

Validate across more weeks before adopting a general winner.

Actual week: 4.820M trips. WAPE sums absolute errors before aggregation; results were unavailable at the planning cutoff.

PROPOSED NEXT DECISION

Test targeted credit on incremental contribution

Proposed experiment · requires internal user, assignment and redemption data

Design element	Operational proposal
Eligibility	Freeze a 28-day pre-test cohort of short / low-fare zone users.
Assignment	Stable user randomization: 90% treatment, 10% holdout.
Primary outcome	Trips and net contribution per assigned user; retain zero-trip users.
Guardrails	Wait time, cancellation rate and contribution per trip.
Duration and power	≥14 issuance days + 7-day follow-up; power from internal data.
Decision	Scale if contribution CI is positive and guardrails hold.

If contribution fails or guardrails worsen: stop or redesign. Public TLC data cannot execute this user-level experiment.

Evidence available for a demand-planning review

From source controls to commercial decisions and a repeatable planning workflow.

Planning capability	Reviewable output
Monitor demand	Four daily segments, watchlist and two documented alert investigations.
Evaluate commercial signals	Fee sensitivity analysis, credit share comparison and a 24-scenario Excel model.
Plan and validate	28 frozen forecasts, an as-of memo and a retrospective backtest.
Communicate decisions	Two report previews, an English planning memo and a proposed credit test.

github.com/handonn2000/urban-mobility-planning

Power BI source draft: 27 measures and two pages. Native DAX, refresh, filters and PBIX validation remain pending.